

Instructor solution

Homework 1 · staff only · fictional coursework

Q1. Solve the system

Augmented matrix $[2 \ 1 \ -1 \ | \ 8; -3 \ -1 \ 2 \ | \ -11; -2 \ 1 \ 2 \ | \ -3]$.

$R1 \leftarrow R1 / 2$. $R2 \leftarrow R2 + 3 R1$. $R3 \leftarrow R3 + 2 R1$.

$R3 \leftarrow R3 - 4 R2$. Back-substitute.

$$\mathbf{x} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$$

Full credit requires every operation labeled and the answer as a column vector.

Q2. Rank and nullity

$\text{rref}(A)$ has two pivots, so $\text{rank } A = 2$.

Nullity $= 3 - 2 = 1$ by rank-nullity.

Null space basis: $\{(-1, 2, 1)^T\}$.

The basis has to be derived from the reduced form, not asserted.

Q3. Subspace proof

Let $u, v \in W$ and $c \in \mathbb{R}$. Then $(u + v)$ satisfies $x + 2y - z = 0$ by linearity.

$(c u)$ satisfies it as well. $0 \in W$ since $0 + 0 - 0 = 0$.

Therefore W is a subspace of $\mathbb{R}^3$.

A pair of worked examples is not a proof; the conclusion must be universal.

Q4. Inverse

$[A \ | \ I] \rightarrow [I \ | \ A^{-1}]$ with labeled operations.

Verification: $A A^{-1} = I$.

$$A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

Cramer or adjugate earn full arithmetic and justification credit.